

# BDH-Spike: A Native Spiking Neural Network Implementation of the Baby Dragon Hatchling Architecture for Neuromorphic Computing with Continual Learning

TakzenAI — Sovereign Neuromorphic Architecture

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## Abstract

The Baby Dragon Hatchling (BDH) architecture replaces attention-softmax machinery with brain-inspired neuron-synapse dynamics, yet its canonical form still operates on continuous floating-point states. We present **BDH-Spike**, a native Spiking Neural Network (SNN) realization of BDH in which every activation inside the core layers is a discrete spike  $S(t) \in \{0, 1\}$ . The framework introduces (i) a Parametric Leaky Integrate-and-Fire neuron with lateral BDH coupling (BDH-PLIF) trained by surrogate gradients, (ii) a softmax-free, multiply-free attention mechanism realized as associative AND-masking on binary spikes with an exact integer bitwise inference path, (iii) a dual-weight plasticity engine separating structural weights ( $W_{\text{slow}}$ , trained offline by BPTT) from episodic weights ( $W_{\text{fast}}$ , updated online by local three-factor Hebbian STDP under **no\_grad**), and (iv) gradient-free homeostatic threshold regulation. On synthetic event-stream classification, BDH-Spike sustains 91–93% temporal sparsity and requires  $9.1\times$  fewer synaptic operations than its dense equivalent. On a five-task split benchmark, episodic  $W_{\text{fast}}$  plasticity reduces measured catastrophic forgetting by  $\sim 24\%$  relative to BPTT-only training. All 123 unit tests pass with strict no-gradient-leak and binary-domain guarantees enforced by construction.

**Keywords:** spiking neural networks; neuromorphic computing; Baby Dragon Hatchling; continual learning; STDP; surrogate gradients; homeostasis; energy-efficient inference

## 1 Introduction

Large sequence models built on transformer machinery dominate machine learning, but their energy profile is fundamentally hostile to edge and neuromorphic hardware: dense multiply-accumulate (MAC) streams over floating-point activations. The Baby Dragon Hatchling (BDH) architecture replaces attention softmax with brain-inspired neuron-synapse dynamics, yet its canonical formulation remains continuous-valued.

We argue that the natural substrate for BDH is the *spiking* regime: neurons integrate evidence over time, communicate through binary events, and update synapses through local plasticity rules rather than global error backpropagation. **BDH-Spike** operationalizes this claim as a complete, tested software stack: a PLIF core with BDH coupling, softmax-free spiking attention, a dual-weight plasticity engine, homeostatic regulation, full model backbones, spike encoders, energy accounting, benchmarks, and diagnostics.

Four architectural invariants govern every module: (1) the *binary spike domain*: all activations in core layers satisfy  $S(t) \in \{0, 1\}$ ; (2) *dual-weight plasticity*: structural weights  $W_{\text{slow}}$  are trained offline via BPTT while episodic weights  $W_{\text{fast}}$  adapt online through local STDP with zero global backpropagation; (3) a *canonical temporal layout*  $[T, B, C, \dots]$  preserved everywhere; and (4) *energy sparsity first*, targeting 85–95% silent states.

## 2 The BDH-PLIF Neuron

The core neuron extends Parametric Leaky Integrate-and-Fire with a lateral BDH recurrent coupling vector  $\mathbf{m}_{\text{BDH}}$  that nonlinearly modulates excitability across time-steps:

$$V[t] = \beta V[t-1] + I_{\text{syn}}[t] + m_{\text{BDH}}[t-1] - S[t-1] V_{\text{th}}, \quad (1)$$

$$S[t] = \Theta(V[t] - V_{\text{th}}), \quad (2)$$

$$m_{\text{BDH}}[t] = \tanh(\rho m_{\text{BDH}}[t-1] + g S[t]), \quad (3)$$

where  $\beta = \sigma(\beta_{\text{logit}})$  is a per-channel learnable decay constrained to  $(0, 1)$ ,  $\rho$  the coupling decay and  $g$  the coupling strength. A hard reset forces  $V[t] \leftarrow 0$  whenever a spike is emitted. Differentiability is provided exclusively by the fast-sigmoid surrogate applied at the threshold crossing,

$$\sigma'(x) = \frac{1}{(1 + k |x - V_{\text{th}}|)^2}, \quad k = 25. \quad (4)$$

Forward values remain strictly binary; floating-point precision exists only to carry gradients during offline training.

## 3 Softmax-Free Spike-Driven Attention

Attention is realized without Softmax, without scaling, and without multiplies. Binary queries, keys and values are produced by thresholding linear projections; association is counted by exact integer accumulation of overlapping spikes; the mask is re-binarized and applied to values:

$$Q_s = \Theta(XW_q - V_{\text{th}}), \quad K_s = \Theta(XW_k - V_{\text{th}}), \quad V_s = \Theta(XW_v - V_{\text{th}}), \quad (5)$$

$$C[n, m] = \sum_c Q_s[n, c] \wedge K_s[m, c], \quad A = \Theta(C - \theta_a), \quad Y = \Theta(AV_s - V_{\text{out}}). \quad (6)$$

Two execution modes share one code path semantics: a *surrogate* mode (binary-valued floats carrying surrogate gradients into  $W_q, W_k, W_v$ ) and a *bitwise* mode producing a strictly boolean/integer graph — boolean AND-mask, exact uint8/int16 accumulation — mapping one-to-one onto POP-COUNT/AC neuromorphic instructions. No float tensor is materialized after input encoding in bitwise mode.

## 4 Dual-Weight Plasticity

Each synapse layer carries two weight matrices with disjoint learning regimes. The fused transmission kernel is

$$I_{\text{syn}} = (W_{\text{slow}} + W_{\text{fast}})^{\top} S_{\text{in}}. \quad (7)$$

$W_{\text{slow}}$  is an ordinary autograd parameter trained offline by BPTT through Eq. (4).  $W_{\text{fast}}$  is a registered buffer updated exclusively inside a `no_grad` region by local three-factor Hebbian (STDP) learning, following the canonical exponential trace recursion

$$A_{\text{pre}}[t] = A_{\text{pre}}[t-1] e^{-\Delta t / \tau_{\text{pre}}} + S_{\text{pre}}[t], \quad (8)$$

$$A_{\text{post}}[t] = A_{\text{post}}[t-1] e^{-\Delta t / \tau_{\text{post}}} + S_{\text{post}}[t], \quad (9)$$

$$\Delta W[t] = M \cdot \left( \eta_+ S_{\text{post}}[t] \otimes A_{\text{pre}}[t] - \eta_- S_{\text{pre}}[t] \otimes A_{\text{post}}[t] \right), \quad (10)$$

where  $M$  is the third (modulatory) factor gating the write into  $W_{\text{fast}}$ , which is clamped to  $|W_{\text{fast}}| \leq w_{\text{max}}$  at every step. Because updates are local and gradient-free by construction (verified by dedicated unit tests), episodic adaptation during inference cannot pollute the autograd graph or erase structural knowledge.

**Homeostasis.** A gradient-free controller keeps network activity inside a healthy band. With a per-layer firing-rate estimate  $r[t] = \beta_r r[t-1] + (1 - \beta_r) \bar{S}[t]$ ,

$$V_{\text{th}}[t] = \text{clip}(V_{\text{th}}[t-1] + \eta_v(r[t] - r_{\text{target}}), V_{\text{min}}, V_{\text{max}}). \quad (11)$$

## 5 Models and Encoders

The stack assembles two backbones. **BDH-Spike-ViT** pipelines images through a convolutional patch projection, min-max normalized steady-rate encoding through the surrogate junction, softmax-free spiking attention, a BDH-PLIF temporal readout over folded tokens, and mean-rate pooling into a linear classifier. **BDH-Spike-Seq** is a streaming block fusing dual-weight synapses with the PLIF cell, supporting online STDP and homeostasis for continuous data. Three event encoders convert real-valued signals into spike trains: rate (Bernoulli), latency (time-to-first-spike), and delta (event-camera style ON/OFF change detection).

## 6 Energy Accounting: SOPs vs. FLOPs

Synaptic operations scale with active spikes rather than tensor density:

$$\text{SOPs} = \sum_{t=1}^T \text{nnz}(S_{\text{in}}[t]) \times \text{FanOut}, \quad (12)$$

against a dense equivalent of  $2TB$  FanIn FanOut FLOPs. Table 1 contrasts the regimes; Table 2 reports measured results.

Table 1: Dense ANN vs. BDH-Spike inference profile.

|                         | Dense ANN                 | BDH-Spike                         |
|-------------------------|---------------------------|-----------------------------------|
| Core operation          | MAC (32-bit float)        | AC / POPCNT (bitwise)             |
| Activation cost         | every neuron, every step  | emitting neurons only             |
| Memory traffic          | full weight matrix / step | active rows only                  |
| Inference-time learning | none                      | online STDP ( $W_{\text{fast}}$ ) |
| Target hardware         | GPU / TPU                 | neuromorphic / edge MCU           |

## 7 Experiments

### 7.1 Event-Stream Classification

On synthetic class-prototyped event streams ( $T=8$ , 5 classes, batch 64), three epochs of surrogate-gradient training yield the energy profile of Table 2. Temporal sparsity stays inside the target band throughout evaluation while accuracy grows monotonically.

Table 2: Measured energy metrics (synthetic event benchmark, epoch 3).

| Quantity                | Measured value      |
|-------------------------|---------------------|
| Temporal spike sparsity | 91–93%              |
| SOPs (evaluation pass)  | $5.52 \times 10^5$  |
| Dense-equivalent FLOPs  | $5.02 \times 10^6$  |
| FLOPs/SOP ratio         | <b>9.1</b> $\times$ |

### 7.2 Split-Task Continual Learning

Five sequential tasks share one hidden layer — the standard condition for catastrophic forgetting. Both schedules train the hidden features identically by SGD; the dual-weight schedule additionally adapts  $W_{\text{fast}}$  online via Eq. (10) while each task stream flows through at inference time. Forgetting is measured as the mean peak-to-final accuracy drop over all tasks except the last:

$$F = \frac{1}{K-1} \sum_{k=1}^{K-1} \left( \max_j \text{acc}_k[j] - \text{acc}_k[K] \right). \quad (13)$$

Dual-weight plasticity reduces forgetting from 0.301 to 0.230 ( $\sim 24\%$  relative reduction) under identical structural training.

## 8 Software Guarantees

The implementation enforces its invariants mechanically: 123 unit tests assert binary-domain outputs, exact  $[T, B, C]$  layouts, hard-reset semantics, surrogate correctness against the closed form of Eq. (4),  $W_{\text{fast}}$  exclusion from `parameters()` with no gradient leak under enabled grad mode, trace decay matching  $e^{-\Delta t/\tau}$  exactly, and weight clamping saturation. All tests pass in under six seconds on CPU.

## 9 Conclusion

BDH-Spike demonstrates that the Baby Dragon Hatchling architecture can be realized natively in the spiking regime without sacrificing trainability or continual adaptability: surrogate gradients preserve BPTT trainability, a bitwise attention path targets neuromorphic silicon directly, and dual-weight plasticity plus homeostasis deliver measurable continual-learning benefits at 91–93% activity sparsity. Future work includes Loihi 2/Lava export, on-chip POPCOUNT kernels, and large-scale event-vision benchmarks.

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